

Exam 3

Andrew Koren

The 2nd paragraph mentions that the complex multiplet dimension M isn't necessarily equal to the dimension N of the symmetry group G . The next paragraph mentions the generators $\hat{T}_n$ indexed $n = 1, 2, \dots N$. I believe this means each $\hat{T}_n$ can be an $M \times M$ matrix or equivalent representation acting on the multiplet. The vector multiplet A_μ^n has the same dimensions as the number of $\hat{T}_n$ with each component being M dimensional.

The infinitesimal transformation $\delta\phi$ is then

$$\begin{aligned}\phi + \delta\phi &= \hat{U}(\delta\omega_n(x))\phi, \\ &= e^{-i\hat{T}_n(\delta\omega_n)}\phi, \\ &\approx (1 - i\hat{T}_n(\delta\omega_n))\phi, \\ \delta\phi &= -i\hat{T}_n(\delta\omega_n)\phi, \\ \delta\phi^\dagger &= i(\delta\omega_n)\phi^\dagger T_n,\end{aligned}$$

where $\delta\omega_n$ is position dependent $\delta\omega_n = \delta\omega_n(x)$. I'm not going to pretend like I can derive $\delta A_\mu^n = C^{nmk}\delta\omega^m A_\mu^k + \frac{1}{g}\partial_\mu\delta\omega^n$, but I do know the group structure constant identity

$$[\hat{T}_n, \hat{T}_m] = C^{nmk}\hat{T}_k.$$

and the generators are hermitian $\hat{T}_k = \hat{T}_k^\dagger$. Local gauge invariance implies $\mathcal{L}$ is conserved, so we will show

$$\delta\mathcal{L} = \frac{1}{2}\delta\left[\underbrace{\mathcal{D}_\mu\phi^\dagger \cdot \mathcal{D}^\mu\phi}_{\text{kinetic term}} - \underbrace{m^2\phi^\dagger \cdot \phi}_{\text{mass term}}\right] = 0$$

The mass term comes easily from the product rule. The mass term is unaffected, yielding

$$\begin{aligned}\delta\left[\phi^\dagger \cdot \phi\right] &= \delta\phi^\dagger \cdot \phi + \phi^\dagger \cdot \delta\phi, \\ &= i\delta\omega_n\left(\phi^\dagger\hat{T}_n\phi - \phi^\dagger\hat{T}_n\phi\right) = 0.\end{aligned}$$

For the kinetic term, we have

$$\delta \left[\mathcal{D}_\mu \phi^\dagger \cdot \mathcal{D}^\mu \phi \right] = \delta \left(\mathcal{D}_\mu \phi^\dagger \right) \mathcal{D}^\mu \phi + \mathcal{D}_\mu \phi^\dagger \delta \left(\mathcal{D}^\mu \phi \right).$$

$$\begin{aligned} \delta \left(\mathcal{D}_\mu \phi \right) &= \partial_\mu \left(\delta \phi \right) + ig \hat{T}_n \left(\delta A_\mu^n \right) \phi + ig \hat{T}_n A_\mu^n \left(\delta \phi \right), \\ &= \partial_\mu \left(-i \hat{T}_n \delta \omega_n \phi \right) + ig \hat{T}_n \left(C^{mnk} \delta \omega_m A_\mu^n + \frac{1}{g} \partial_\mu \delta \omega_n \right) \phi + ig \hat{T}_n A_\mu^n \left(-i \hat{T}_{n'} \delta \omega_{n'} \phi \right), \\ &= -i \cancel{\partial_\mu \delta \omega_n \hat{T}_n \phi} - i \delta \omega_n \hat{T}_n \partial_\mu \phi + ig C^{mnk} \delta \omega_m A_\mu^k \hat{T}_n \phi + i \cancel{\partial_\mu \delta \omega_n \hat{T}_n \phi} + g \delta \omega_{n'} A_\mu^n \hat{T}_n \hat{T}_{n'} \\ &= -i \delta \omega_n \hat{T}_n \partial_\mu \phi + ig \delta \omega_m A_\mu^k C^{mnk} \hat{T}_n \phi + g \underbrace{\delta \omega_{n'}}_{n' \rightarrow k} \underbrace{A_\mu^n}_{n \rightarrow m} \underbrace{\hat{T}_n}_{n \rightarrow m} \underbrace{\hat{T}_{n'}}_{n' \rightarrow k} \phi, \\ &= -i \delta \omega_n \hat{T}_n \partial_\mu \phi + g \delta \omega_m A_\mu^k \left(i C^{mnk} \hat{T}_n + \hat{T}_k \hat{T}_m \right) \phi, \\ &= -i \delta \omega_n \hat{T}_n \partial_\mu \phi + \underbrace{g \delta \omega_m A_\mu^k \hat{T}_m \hat{T}_k}_{m \rightarrow n} \phi, \\ &= -i \delta \omega_n \hat{T}_n \mathcal{D}_\mu \phi. \end{aligned}$$

We should be able to make some argument about the change in the adjoint $\delta \left(\mathcal{D}_\mu \phi^\dagger \right)$ without having to do too much math, right? In this context $\mathcal{D}_\mu \phi^\dagger \equiv \left(\mathcal{D}_\mu \phi \right)^\dagger$, so we should have $\delta \mathcal{D}_\mu \phi^\dagger = i \delta \omega_n \mathcal{D}_\mu \phi^\dagger \hat{T}_n$.

$$\delta \left(\mathcal{D}_\mu \phi^\dagger \right) \mathcal{D}^\mu \phi \mathcal{D}_\mu \phi^\dagger \delta \left(\mathcal{D}^\mu \phi \right) = i \delta \omega_n \mathcal{D}_\mu \phi^\dagger \hat{T}_n \mathcal{D}^\mu \phi - i \delta \omega_n \mathcal{D}_\mu \phi^\dagger \hat{T}_n \mathcal{D}^\mu \phi = 0.$$

Together yielding

$$\delta \mathcal{L} = 0.$$

Switching from ∂_μ to $\mathcal{D}_\mu$ actually says this lagrangian does vary, just that that variation is captured by the variation in A_μ^n .

Vector field strength tensor $F_{\mu\nu}^n$. Just expand out the commutator.

$$[\mathcal{D}_\mu, \mathcal{D}_\nu] \phi = \mathcal{D}_\mu \mathcal{D}_\nu \phi - \mathcal{D}_\nu \mathcal{D}_\mu \phi$$

Well, we still have to get that $ig \hat{T}_n$ off of the tensor.

$$\begin{aligned} \mathcal{D}_\mu \mathcal{D}_\nu \phi &= \left(\partial_\mu + ig \hat{T}_n A_\mu^n \right) \left(\partial_\nu + ig \hat{T}_n A_\nu^n \right) \phi, \\ &= \partial_\mu \partial_\nu \phi + \partial_\mu \left(ig \hat{T}_n A_\nu^n \phi \right) + ig \hat{T}_n A_\mu^n \partial_\nu \phi - g^2 A_\mu^n A_\nu^{n'} \hat{T}_n \hat{T}_{n'} \phi, \\ &= \partial_\mu \partial_\nu \phi + ig \hat{T}_n \left(\partial_\mu A_\nu^n \right) \phi + ig \hat{T}_n A_\nu^n \partial_\mu \phi + ig \hat{T}_n A_\mu^n \partial_\nu \phi - g^2 A_\mu^n A_\nu^{n'} \hat{T}_n \hat{T}_{n'} \phi. \\ [\mathcal{D}_\mu, \mathcal{D}_\nu] \phi &= \partial_\mu \partial_\nu \phi + ig \hat{T}_n \left(\partial_\mu A_\nu^n \right) \phi + \cancel{ig \hat{T}_n A_\nu^n \partial_\mu \phi} + \cancel{ig \hat{T}_n A_\mu^n \partial_\nu \phi} - g^2 A_\mu^n A_\nu^{n'} \hat{T}_n \hat{T}_{n'} \phi \\ &\quad - \partial_\nu \partial_\mu \phi - ig \hat{T}_n \left(\partial_\nu A_\mu^n \right) \phi - \cancel{ig \hat{T}_n A_\mu^n \partial_\nu \phi} - \cancel{ig \hat{T}_n A_\nu^n \partial_\mu \phi} + g^2 A_\mu^{n'} A_\nu^n \hat{T}_n \hat{T}_{n'} \phi. \end{aligned}$$

We'll have to combine the double $\hat{T}_n$ terms and the double derivative must go to zero

$$\begin{aligned}
& \partial_\mu \partial_\nu \phi - \partial_\nu \partial_\mu \phi = 0, \\
& -g^2 A_\mu^n A_\nu^{n'} \hat{T}_n \hat{T}_{n'} \phi + g^2 A_\mu^{n'} A_\nu^n \hat{T}_n \hat{T}_{n'} \phi = g^2 A_\mu^n A_\nu^{n'} \left(\underbrace{\hat{T}_{n'}}_{n \rightarrow n' n' \rightarrow n} \hat{T}_n - \hat{T}_n \hat{T}_{n'} \right) \phi, \\
& = -ig^2 A_\mu^m A_\nu^k C^{nmk} \hat{T}_n \phi.
\end{aligned}$$

Combining the remaining like terms yields

$$\begin{aligned}
[\mathcal{D}_\mu, \mathcal{D}_\nu] \phi &= ig \hat{T}_n [\partial_\mu A_\nu^n - \partial_\nu A_\mu^n - g C^{mmk} A_\mu^m A_\nu^k] \phi \\
F_{\mu\nu}^n &= [\partial_\mu A_\nu^n - \partial_\nu A_\mu^n - g C^{mmk} A_\mu^m A_\nu^k] \phi
\end{aligned}$$

although I think we dropped the ϕ . How is $F_{\mu\nu}^n$ transformed? Thanks to linearity, the commutator transforms the same way as its operators and we have

$$\delta([\mathcal{D}_\mu, \mathcal{D}_\nu] \phi) = -i \hat{T}_n \delta \omega_n [\mathcal{D}_\mu, \mathcal{D}_\nu] \phi,$$

so we only need to expand this expression and dig out $\delta F_{\mu\nu}^n$. Each index below is paired, so reindexing freely we have

$$\begin{aligned}
-i \hat{T}_{n'} \delta \omega_{n'} [\mathcal{D}_\mu, \mathcal{D}_\nu] \phi &= -i \hat{T}_{n'} \delta \omega_{n'} (ig \hat{T}_n F_{\mu\nu}^n) \phi = g \delta \omega_{n'} F_{\mu\nu}^n \hat{T}_{n'} \hat{T}_n \phi, \\
\delta (ig \hat{T}_n F_{\mu\nu}^n \phi) &= ig \hat{T}_n (\delta F_{\mu\nu}^n) \phi + ig \hat{T}_n F_{\mu\nu}^n \delta \phi, \\
&= ig \hat{T}_n (\delta F_{\mu\nu}^n) \phi + g F_{\mu\nu}^n \delta \omega_{n'} \hat{T}_n \hat{T}_{n'} \phi, \\
i \hat{T}_n (\delta F_{\mu\nu}^n) &= -\delta \omega_{n'} F_{\mu\nu}^n \hat{T}_n \hat{T}_{n'} + \delta \omega_{n'} F_{\mu\nu}^n \hat{T}_{n'} \hat{T}_n, \\
&= \delta \omega_{n'} F_{\mu\nu}^n [\hat{T}_{n'}, \hat{T}_n] = \delta \omega_m F_{\mu\nu}^k [\hat{T}_m, \hat{T}_k], \\
&= i \hat{T}_n C^{mmk} \delta \omega_m F_{\mu\nu}^k,
\end{aligned}$$

yielding our result

$$\delta F_{\mu\nu}^n = C^{mmk} \delta \omega_m F_{\mu\nu}^k.$$

Naturally, we arrive at the invariant Lagrangian

$$\begin{aligned}
\mathcal{L}_{YM} &= -\frac{1}{4} F_{\mu\nu}^n F^{n\mu\nu}, \\
-4\delta \mathcal{L}_{YM} &= (\delta F_{\mu\nu}^n) F^{n\mu\nu} + F_{\mu\nu}^n \delta F^{n\mu\nu}, \\
&= C^{mmk} \delta \omega_m F_{\mu\nu}^k F^{n\mu\nu} + C^{mmk} \delta \omega_m F_{\mu\nu}^n F^{k\mu\nu}, \\
&= C^{mmk} \delta \omega_m F_{\mu\nu}^k F^{n\mu\nu} - C^{mmk} \delta \omega_m F_{\mu\nu}^k F^{n\mu\nu} = 0.
\end{aligned}$$

It looks like the Yang-Mills paper used $\hat{T}_n = \sigma_n$ instead of $\frac{\sigma_n}{2}$, which we would get for

$$\begin{aligned}
f_{\mu\nu} &= \partial_\nu b_\mu - \partial_\mu b_\nu - 2g b_\mu \times b_\nu \\
f_{\mu\nu}^n &= \partial_\nu b_\mu^n - \partial_\mu b_\nu^n - 2g C^{nmk} b_\mu^m b_\nu^k \\
&= \partial_\nu b_\mu^n - \partial_\mu b_\nu^n - 2g C^{nmk} b_\mu^m b_\nu^k
\end{aligned}$$

For $SU(2)$ we use the pauli matrices and levi-civita symbol

$$\begin{aligned}
C^{nmk} &= \epsilon^{nmk} \\
\hat{T}_n &= \frac{\sigma_n}{2} \\
[\hat{T}_m, \hat{T}_k] &= i\epsilon^{nmk} T_n
\end{aligned}$$

But I think we need the mapping $A_\mu^n = -2b_\mu^n$, so the original paper probably used $\hat{T}_n = \sigma_n$. No idea about the minus sign

$$\begin{aligned}
F_{\mu\nu}^n &= \partial_\mu A_\nu - \partial_\nu A_\mu - g\epsilon^{nmk} A_\mu^m A_\nu^k \\
&= -2(\partial_\mu b_\nu^n - \partial_\nu b_\mu^n + 2g C^{nmk} b_\mu^m b_\nu^k) \\
&= 2(\partial_\nu b_\mu^n - \partial_\mu b_\nu^n - 2g C^{nmk} b_\mu^m b_\nu^k) \\
&= 2f_{\mu\nu}^n \\
F_{\mu\nu} &= 2f_{\mu\nu}
\end{aligned}$$